LAB 6: CODE COVERAGE ANALYSIS

Gym Membership Application

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Course: BTech – CSE (AI & ML)

Assigned Repository: L6_L7_21_gym_membership

GitHub Repository: https://github.com/Preeti1k3/L6_L7_21_gym_membership

Tools Used: Python 3.13.1, Pytest 8.3.4 and Coverage.py

1.Aim:

To measure the percentage of executable statements in the Gym Membership application that are executed by the test cases using Pytest and Coverage.py, and to improve statement coverage to at least 90%.

2. Objective

The objective is to identify statements that are not executed by the starter tests, write additional test cases to exercise the application functions, and analyse the resulting code coverage report.

3. Application Under Test

The application file gym_membership.py contains three functions. membership_cost() calculates membership fees using the plan, age, student status and trainer add-on. can_enter() checks gym entry using membership status, operating hours and women-only slot rules. bmi_category() calculates BMI and returns a weight category.

4. Initial Test Execution and Analysis

The starter test file test_gym_membership.py contained two tests: test_monthly_plan() and test_can_enter(). These tests covered a basic monthly membership calculation and a normal gym entry case.

Command used:

```
coverage run --branch --source=gym_membership -m pytest -v
test_gym_membership.py
```

The initial coverage report showed 35 statements, 20 missed statements and 38% coverage. Both starter tests passed. The limited test set did not exercise many application behaviours, including invalid plans, age restrictions, discounts, trainer charges, entry restrictions and BMI categories.

Note: The 38% figure is the initial coverage value shown in the retained initial report; it is included here as the baseline result.

5. Test Cases Added for Code Coverage

A separate test file named CodeCoverageTest_21_gym_membership.py was created. The 20 test cases exercised the following application behaviours:

- Monthly, quarterly and yearly membership plans.
- Invalid membership plan and age below the minimum allowed age.
- Senior citizen discount and student discount.
- Trainer add-on and yearly-plan trainer adjustment.
- Inactive membership and gym opening/closing restrictions.
- Female and male entry during women-only slots, and regular-slot entry.
- Invalid height and the underweight, normal, overweight and obese BMI categories.

6. Code Coverage Command

```
coverage run --source=gym_membership -m pytest -v
CodeCoverageTest_21_gym_membership.py
```

```
coverage report -m
```

7. Code Coverage Results

Metric	Result
Tests passed	20

Total statements 35

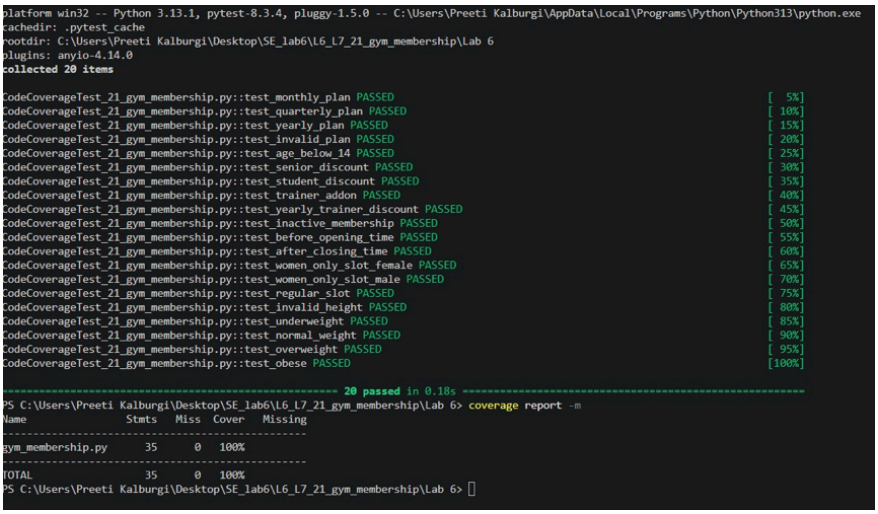
Statements missed 0

Code coverage 100%

Analysis: All 20 tests passed. The report showed that all 35 executable statements in gym_membership.py were executed and no statements were missed. The resulting code coverage was 100%, exceeding the required minimum of 90%.

8. Terminal Output Screenshot

Code coverage test execution and coverage report:



9. Git Commit

Commit ID: 9fe446d
Commit message: Add tests achieving 100 percent code coverage
The code coverage test file and its screenshot were committed and pushed to the personal GitHub fork.

10. Conclusion

Code coverage analysis was performed on the Gym Membership application. The starter tests provided limited coverage, while the additional 20 test cases exercised all 35 executable statements. The final code coverage achieved was 100%, satisfying and exceeding the lab requirement.